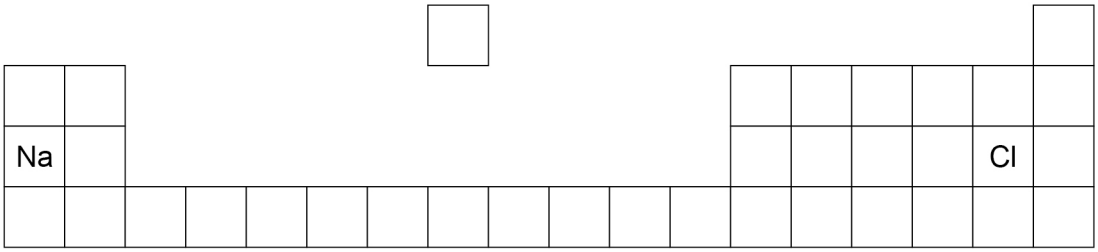
Do not write outside the box

**0 2** This question is about sodium and chlorine.

**Figure 2** shows the positions of sodium and chlorine in the periodic table.

**Figure 2**




**0 2 . 1** State **one** difference and **one** similarity in the electronic structure of sodium and of chlorine.

**[2 marks]**

Difference _____

_____

Similarity _____

_____

**0 2 . 2** Sodium atoms react with chlorine atoms to produce sodium chloride (NaCl).

Describe what happens when a sodium atom reacts with a chlorine atom.

Write about electron transfer in your answer.

**[4 marks]**

_____

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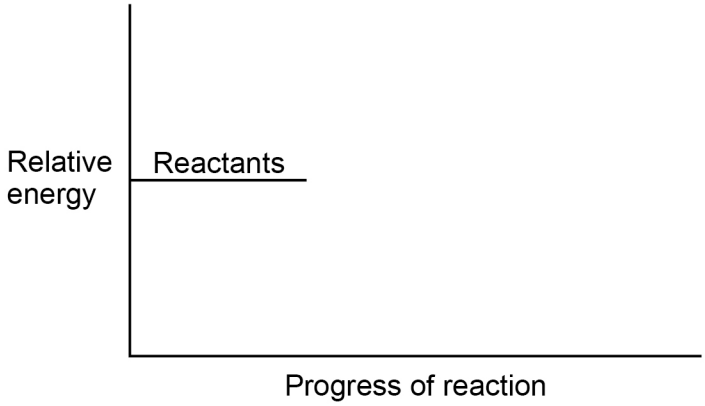
**0 2 . 3** The reaction between sodium and chlorine is an exothermic reaction.

Complete the reaction profile for the reaction between sodium and chlorine.

**[2 marks]**

**Figure 3**




8

Do not write outside the box

**0 3** A student plans a method to prepare pure crystals of copper sulfate.

The student's method is:

1. Add one spatula of calcium carbonate to dilute hydrochloric acid in a beaker.
2. When the fizzing stops, heat the solution with a Bunsen burner until all the liquid is gone.

The method contains several errors and does not produce copper sulfate crystals.

Explain the improvements the student should make to the method so that pure crystals of copper sulfate are produced.

**[6 marks]**

6